

Palindrome Number

Company: Deloitte

Round: Offline

Year: 2022

Difficulty: Easy

Topic: Numbers, Loops, Mathematics

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Problem Description

Given an integer N, check whether the number is a palindrome.

A number is a palindrome if it remains the same when its digits are reversed.

Example

N = 121

Reverse of 121 is also:

121

Therefore, it is a palindrome.

Input Format

The input contains a single integer N.

Output Format

- Print true if N is a palindrome.
- Otherwise, print false.

Sample Input

121

Sample Output

True

Explanation

A palindrome number is a number that remains the same when its digits are reversed.

For example:

121 $\rightarrow$ 121

So, 121 is a palindrome.

How it works

For the input:

121

We reverse the number digit by digit:

121 $\rightarrow$ 12 $\rightarrow$ 1 $\rightarrow$ 0

The digits are taken from the last to the first:

$1 \rightarrow 2 \rightarrow 1$

So the reversed number becomes:

121

Now compare:

Original = 121

Reversed = 121

Since both are equal, the answer is:

true

Approach to Solve This Problem:

1. Store the original number.
2. Reverse the number digit by digit.
3. Compare the reversed number with the original number.
4. If both are equal, return true.
5. Otherwise, return false.

Java Solution:

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int original = n;
7         int reverse = 0;
8         while (n != 0) {
9             int digit = n % 10;
10            reverse = reverse * 10 + digit;
11            n = n / 10;
12        }
13        if (original == reverse) {
14            System.out.println("true");
15        } else {
16            System.out.println("false");
17        }
18        sc.close();
19    }
20 }
```

Solution:

The solution is to reverse the given number and compare it with the original number.

1. Store the original number in a variable.
2. Extract the last digit using:

$\text{digit} = n \% 10$

3. Add this digit to the reversed number.
4. Remove the last digit using:

$n = n / 10$

5. Repeat until all digits are processed.
6. Compare the reversed number with the original number.
7. If both are equal, print true; otherwise, print false.

Example

For:

$N = 121$

Reverse:

$121 \rightarrow 121$

Since:

Original = 121

Reverse = 121

the answer is:

true

Complexity Analysis:

Time Complexity: $O(\log N)$

- The algorithm processes the number **digit by digit**.
- A number N has approximately $\log_{10}(N)$ digits.
- Each digit is processed exactly once while reversing the number.

Space Complexity: $O(1)$

- Only a few variables are used, such as original, reverse, digit, and n.
- No extra data structures are required.